8 **Semiconductors and Light-Emitting Diodes (LEDs)**

The following article describes how semiconductors are made into devices such as diodes, light emitting diodes and quantum-well light emitting diodes.

Doped Silicon Semiconductors

Some semiconductors are made of silicon which is an electrical insulator. To allow silicon to conduct electricity, a process known as doping adds impurity atoms into the silicon crystal lattice to increase the number density of charge carriers. Impurity atoms such as phosphorus create an **n-type semiconductor** where **free electrons** are the majority charge carriers. On the other hand, doping with atoms such as boron creates a **p-type semiconductor** where the majority charge carriers are **positive holes**.

A sample of an n-type silicon semiconductor is shown in Fig. 8.1. together with its dimensions.

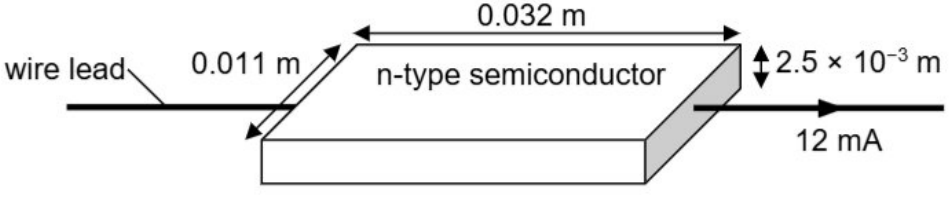


Fig. 8.1

Pure silicon has a density of 2.33×10^3 kg m^{-3} and a molar mass of 28.1×10^{-3} kg mol^{-1}.

The n-type silicon slab in Fig. 8.1 was doped with **one phosphorus atom for every 2.0×10^6 silicon atoms.** Each phosphorus atom donates one free conduction electron. In the sample above, the doping in the n-type silicon slab allows it to carry a current of 12 mA.

The p-n Junction Diode

Something interesting happens when you dope half of a silicon slab with n-type atoms and the other half with p-type atoms: you form a **p-n junction diode**.

Fig. 8.2 shows a diagram of a p-n junction diode.

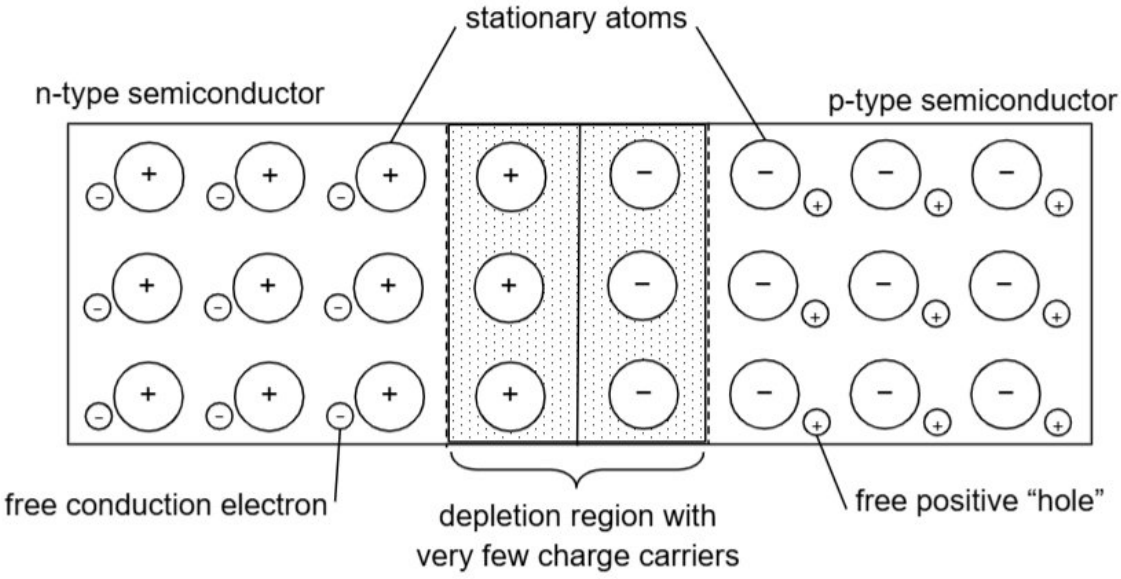


Fig. 8.2

In the central shaded region, known as the **depletion region,** the mobile electrons and holes recombine and neutralise one another, so that the shaded region has very few mobile charge carriers. A potential gradient is set up in the region, which creates an electric field that opposes the diffusion of charge carriers. The potential gradient in the depletion region of width 0.33 μm is shown in Fig. 8.3.

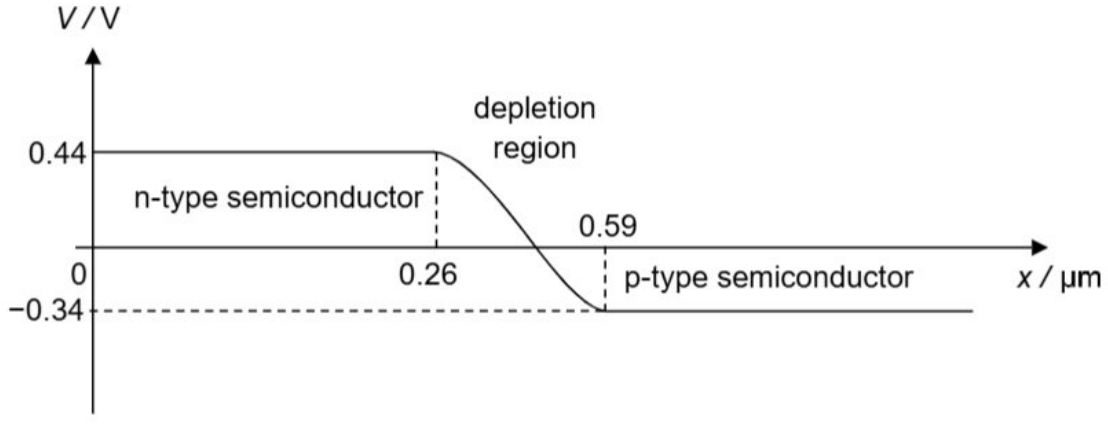


Fig. 8.3

Light-Emitting Diodes (LEDs)

Some semiconductor diodes such as Gallium Arsenide (GaAs) were found to produce light when conducting current in forward-bias mode. In these light-emitting diodes (LEDs), when electrons and holes recombine, a higher energy level electron in the conduction band will transition to a lower energy level hole in the valence band and release energy in the form of a photon.

An example of the energy levels in a GaAs LED is shown in Fig. 8.4.

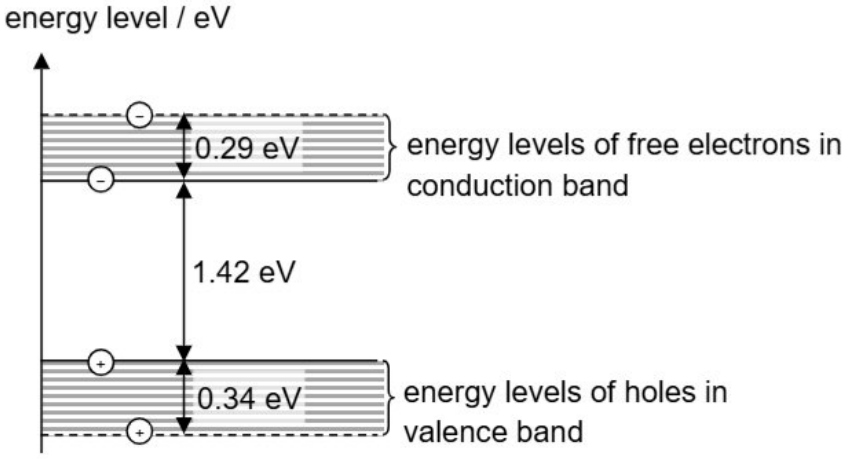


Fig. 8.4

The free electrons in the conduction band occupy a range of energy levels, as do the holes in the valence band. As a result, light emitted from LEDs is not monochromatic and is not a “pure” colour.

Quantum-Well Light Emitting Diodes (QLEDs)

The broad emission spectrum of conventional LEDs makes them unsuitable for use in devices requiring pure colors, such as in high-definition display screens. To address this shortcoming, modern display technologies employ **Quantum-Well Light-Emitting Diodes (QLEDs)**.

A quantum well is made by sandwiching a very thin layer of a semiconductor between thicker layers of another type of semiconductor. When the width of this central layer is reduced to near the de Broglie wavelength of an electron (around a few nanometers thick), quantum effects in the system become significant.

An electron trapped inside this ultra-thin semiconductor layer can be modeled as a particle trapped within a one-dimensional potential energy well, bounded by infinite potential walls. This strict confinement forces the electron's energy to be restricted to a few discrete values. A monitor screen made from such QLED pixels can thus show pictures with exceptionally pure colours.

(a) Show that the number density of free electrons in the n-type semiconductor in Fig. 8.1 is 2.5×10^{22} m^{-3}.

[2]

(b) Calculate the drift velocity of the electrons in the n-type semiconductor in Fig. 8.1.

drift velocity = ... $m\ s^{-1}$ [2]

(c) With reference to the electric potential graph in Fig. 8.3, explain the direction of the electric field in the depletion region.

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..... [2]

(d) Using Fig. 8.3, calculate the average electric field strength within the depletion region.

electric field strength = ... V m^{-1} [2]

(e) The depletion region has positive and negative charges separated on either side, similar to a parallel plate capacitor. The stored charge in the depletion region is equal on both sides and has a magnitude of 3.6×10^{-8} C.

Estimate the capacitance of the depletion layer.

capacitance = ... F [2]

(f) With reference to Fig. 8.4, determine the minimum wavelength of the photons that could be produced by electron-hole recombination in the GaAs LED.

minimum wavelength = ... nm [2]

(g) State the region of the electromagnetic spectrum in which the wavelength of your answer to **(f)** lies.

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..... [1]

(h) Fig. 8.5 below shows the energy levels in the conduction band of a quantum well semiconductor LED. The quantum well has a thickness d of 5.0×10^{-9} m and a potential energy barrier on either side of the quantum well of $E_b = 0.14$ eV.

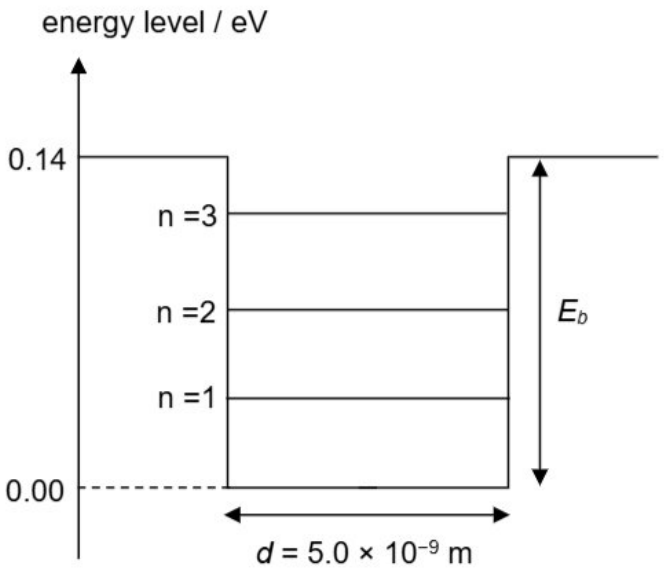


Fig. 8.5

(i) On Fig. 8.5, sketch the wavefunctions that represent the electron in the $n = 1$ and $n = 3$ energy levels.

[2]

(ii) Given that the allowed values of momentum for the electron in the nth energy level of the quantum well is given by

$$p_n = \frac{nh}{2d}$$

show that the allowed values of the energy E_n of the electron in each energy level n is given by the expression

$$E_n = \frac{n^2h^2}{8md^2}$$

[2]

(iii) Determine the energy of the $n = 3$ energy level.

Hence, explain why this is the highest discrete energy level that can be found within the quantum well.

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..... [3]